

Autodesk® Scaleform®

Scaleform 4.2

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1.03

2012 8 2

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Autodesk® Scaleform® 4.2

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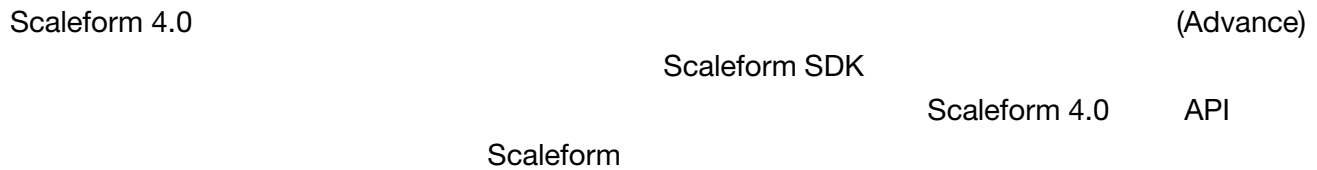
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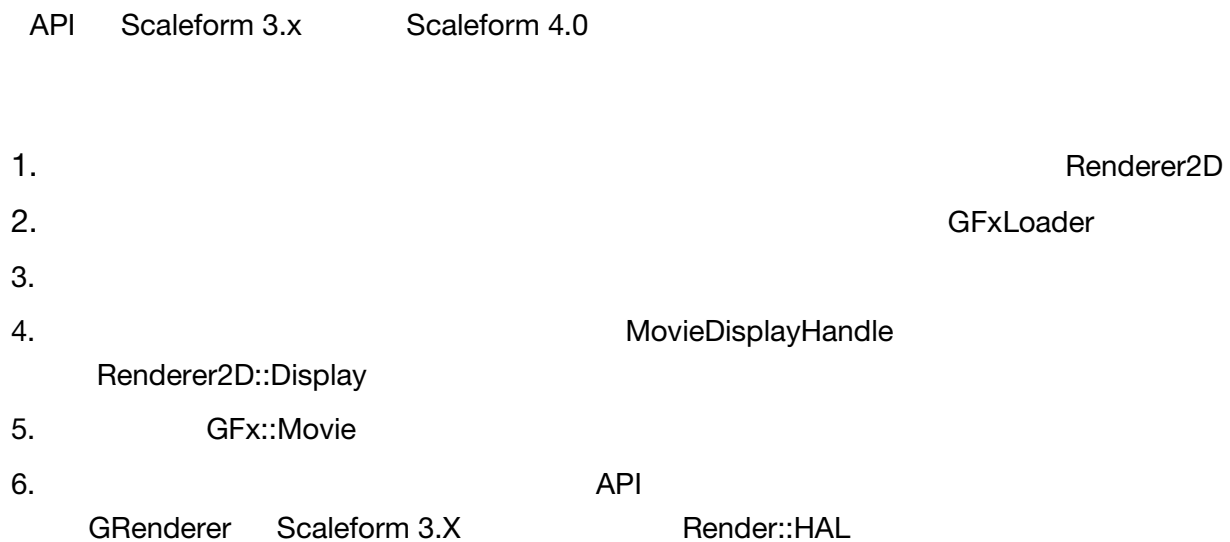
Document	Scaleform 4.2 Renderer Threading Guide
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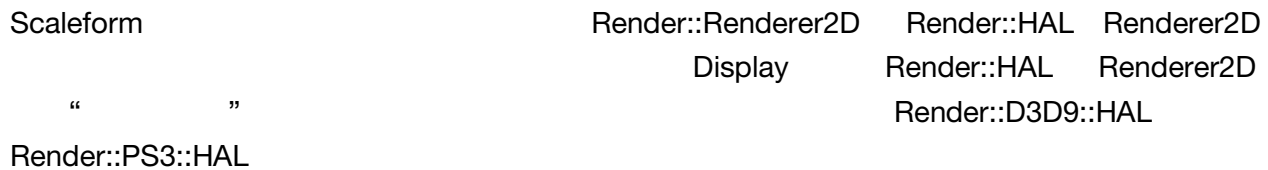
1



2



2.1



```
D3DPRESENT_PARAMETERS PresentParams;  
IDirect3DDevice9*      pDevice = ...;  
ThreadId               renderThreadId = Scaleform::GetCurrentThreadId();  
Render::ThreadCommandQueue *commandQueue = ...;  
Ptr<Render::D3D9::HAL>  pHal;  
Ptr<Render::Renderer2D> pRenderer;  
  
pHal = *SF_NEW Render::D3D9::HAL(commandQueue);
```

```

if (!pHal-> InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
                                                renderThreadId)))
{
    return false;
}
pRenderer = *SF_NEW Render::Renderer2D(pHal);

```

HALInitParams class
 PresentParams
 ThreadCommandQueue
 renderThreadId
 pDevice
 Renderer2D
 HAL

2.2

Scaleform 4.0
 Scaleform 3.x
 Render::HAL
 Renderer2D
 GfxLoader
 ShutdownHAL
 InitHAL
 Render::HAL
 HAL

```

MeshCacheParams mcp;
mcp.MemReserve      = 1024 * 1024 * 3;
mcp.MemLimit        = 1024 * 1024 * 12;
mcp.MemGranularity  = 1024 * 1024 * 3;
mcp.LRUTailSize     = 1024 * 1024 * 4;
mcp.StagingBufferSize = 1024 * 1024 * 2;

pRenderer->GetMeshCacheConfig()->SetParams(mcp);

```

MemReserve
 LRU
 LRUTailSize
 MeshCache
 MeshCache
 64K
 MemLimit
 MemGranularity
 StagingBufferSize
 PC

```

GlyphCacheParams gcp;
gcp.TextureWidth = 1024;
gcp.TextureHeight = 1024;
gcp.NumTextures = 1;
gcp.MaxSlotHeight = 48;

pRenderer->GetGlyphCacheConfig()->SetParams(gcp);

```

Scaleform 3.3	TextureWidth	TextureHeight	NumTextures
	MaxSlotHeight		

InitHAL	InitHAL	SetParams
---------	---------	-----------

2.3

Scaleform 4.0	GfX::ImageCreator	Render::TextureManager
	Scaleform	ImageCreator
		GfX::Loader

```

GfX::Loader loader;
...

SF::Ptr<GfX::ImageCreator> pimageCreator =
    *new GfX::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);

```

ImageCreator	TextureManager	TextureManager
--------------	----------------	----------------

Render::HAL::GetTextureManager()
GetTextureManager

Render::HAL

TextureManager

CreateTexture

-

D3DCREATE_MULTITHREADED

D3D9

-

ThreadCommandQueue

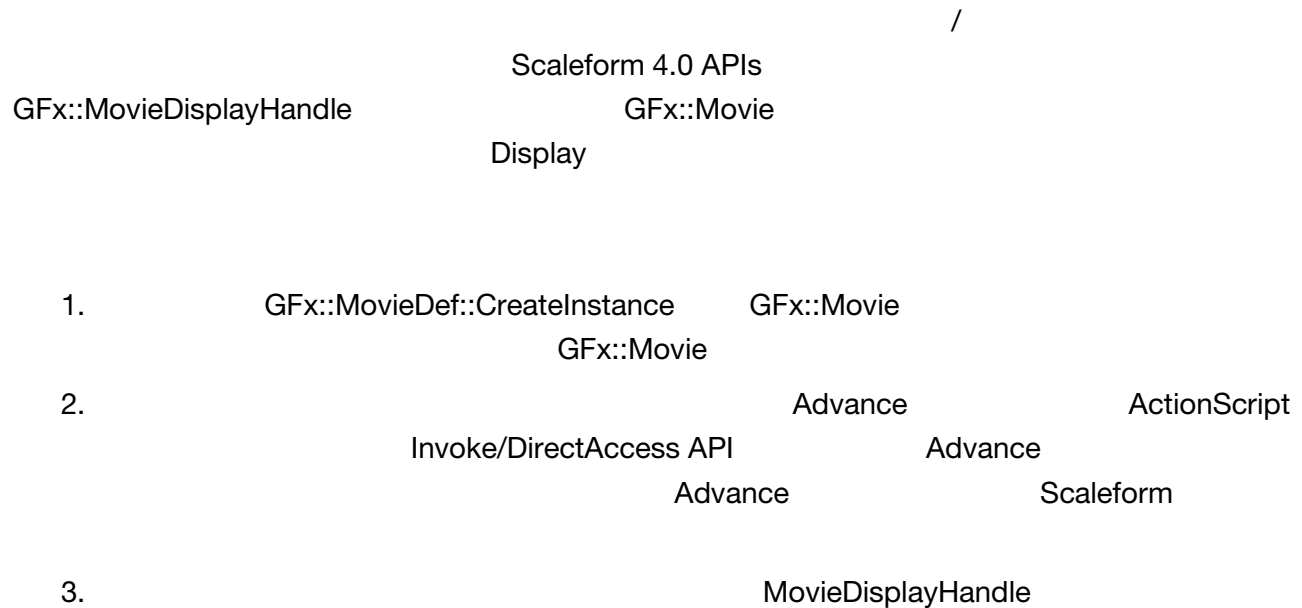
Render::HAL

D3D

ThreadCommandQueue

Platform::RenderHALThread

2.4



```
//-----
// 1.
Ptr<GfX::Movie>      pMovie = ...;
GfX::MovieDisplayHandle hMovieDisplay;

//
//
pMovie->SetViewport(width, height, 0,0, width, height);
hMovieDisplay = pMovie->GetDisplayHandle();

//
//          Scaleform Player
//
pRenderThread->AddDisplayHandle(hMovieDisplay);

//-----
// 2.
//
//  ExternalInterface      Advance
float deltaT = ...;
pMovie->HandleEvent(...);
pMovie->Advance(deltaT);
```

```

//
//
pRenderThread->WaitForOutstandingDrawFrame();
pRenderThread->DrawFrame();

//-----
// 3.         DrawFrame
Ptr<Render::Renderer2D> pRenderer = ...;

pRenderer->BeginFrame();
bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());
if (hasData)
    pRenderer->Display(hMovieDisplay);
pRenderer->EndFrame();

```

- WaitForOutstandingFrame – CPU
- Advance
- MovieDisplay.NextCapture – “ ”Advance false

2.5

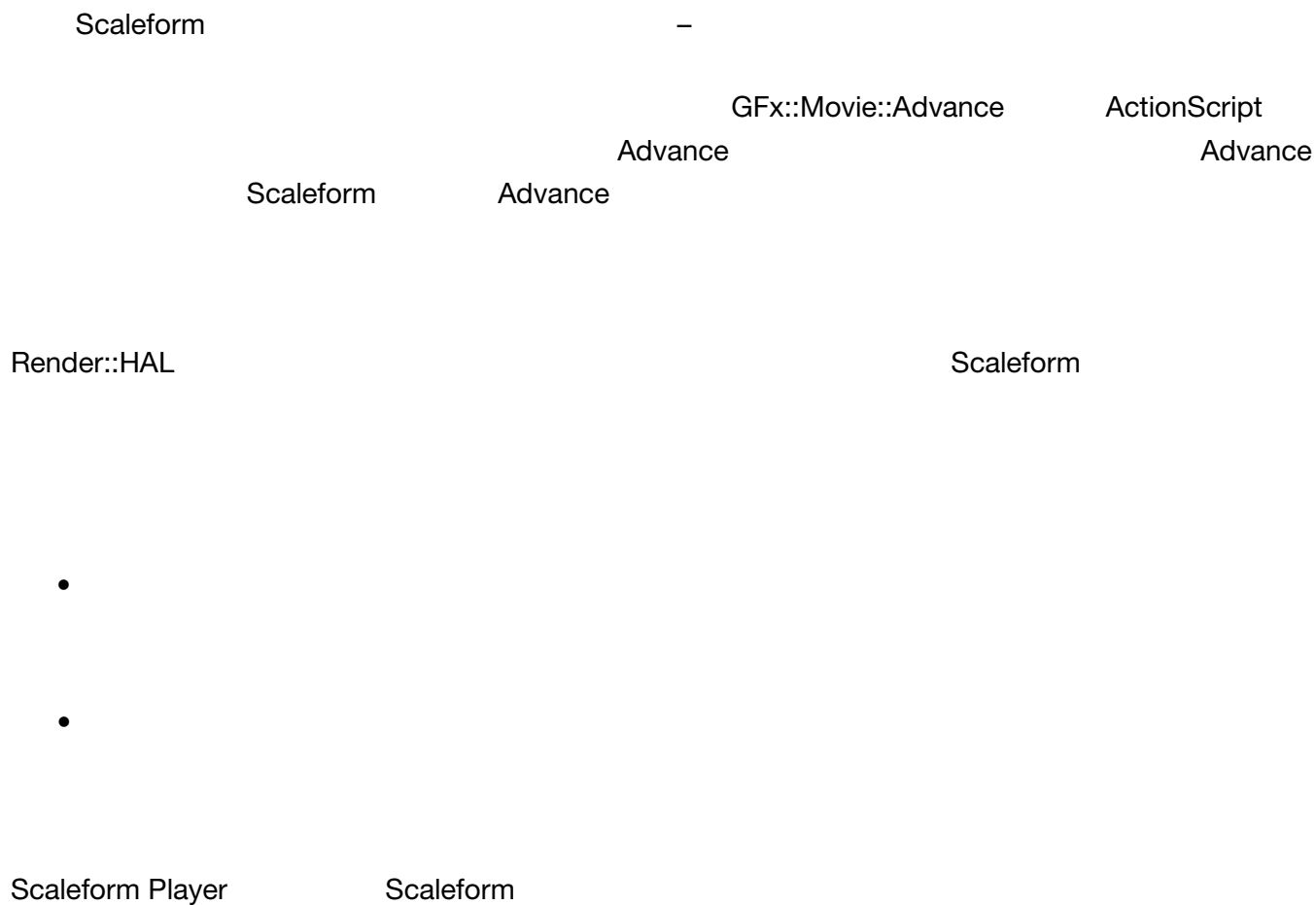
GFx::Movie Release

Render::ThreadCommandQueue Movie
ThreadCommandQueue

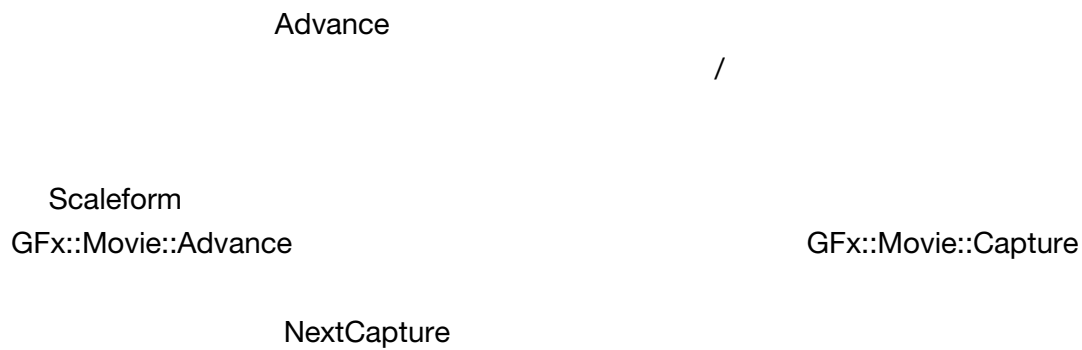
Movie::ShutdownRendering(false)

Movie::IsShutdownRenderingComplete
NextCapture false

3



3.1



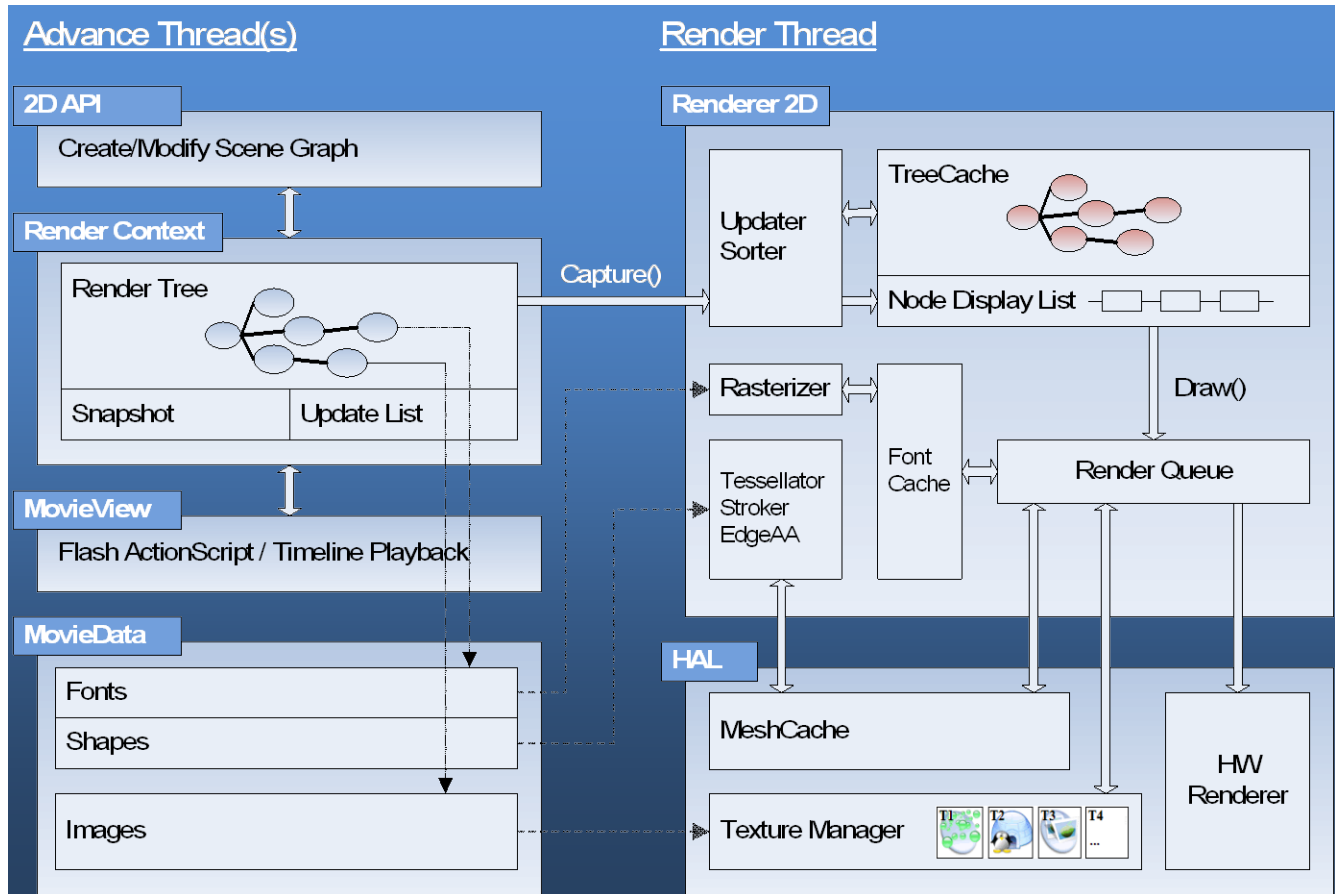
- Movie::Advance
- Capture NextCapture Capture
- Capture Advance Invoke Direct Access API
- Movie

Scaleform 4.0 Scaleform 3.3

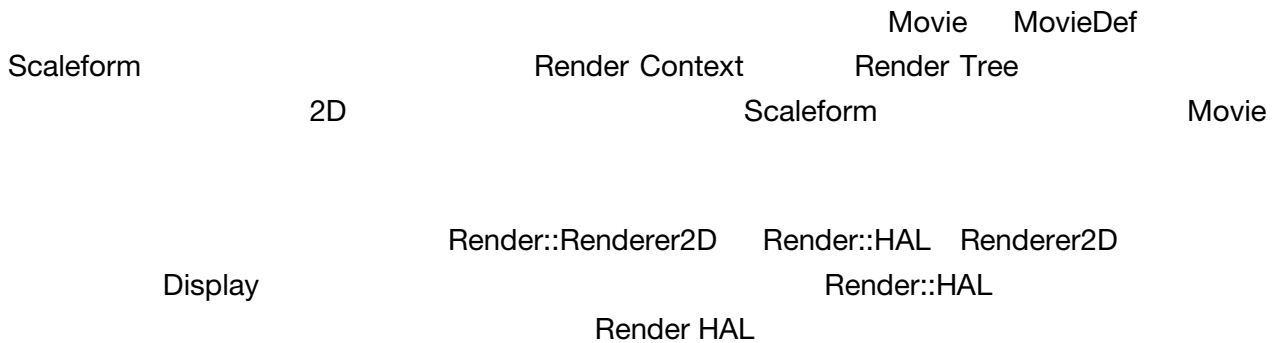
Display Movie::Invoke Capture

Advance Advance Invoke

Scaleform 4.0



1



-
- Font Cache Font Cache
- Mesh Cache
- Render Queue Render Queue /
CPU
- HAL API

ActionScript AI
/

Scaleform

ActionScript

5.2

AS3

Src/Render/ShaderData.xml

“Position2dOffset”

“Position2dOffset”

```
<ShaderSource id="Position2dOffset" pipeline="Vertex">
```

```
    Position2d(
    attribute float4 pos      : POSITION,
    varying   float4 vpos    : POSITION,
    uniform   float4 mvp[2],
    uniform   float  poffset)
    {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
    }
</ShaderSource>
```

“Position2d” – ‘poffset’ x y

D3D9_Shader.cpp ShaderInterface::SetStaticShader

```
//
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
    {
//
        vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
        shader  = (FragShaderDesc::ShaderType) ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
    }
}
```

HAL UserDataStack “abc”

Position2dOffset
D3D9_ShaderDescs.h ShaderData.xml

‘poffset’

ShaderInterface::Finish

```
//
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
```



```

        if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
        {
//      poffset
            for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
            {
                float poffset = data->RendererFloat / 256.0f;
                SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
            }
        }
    }
}

```

```

        'meshCount'
meshCount      1      HAL      'm'

```

5.3

```

HAL      Scaleform

Scaleform      /      Scaleform      AS2
"disableBatching"      AS3 "disableBatching"      AS2/AS3

```

6 GFxShaderMaker

6.1

Scaleform 4.1 GFxShaderMaker Scaleform
 Render_D3D9 ShaderData.xml
/

D3D9

Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp

6.2 *API HAL*

6.2.1 D3D9

D3D9 HAL ShaderModel 2.0 ShaderModel 3.0 HAL

GFxShaderMaker '-shadermodel'

GFxShaderMaker.exe -platform D3D1x -shadermodel SM30

 ShaderModel 2.0 ShaderModel 2.0

InitHAL

6.2.2 D3D1x

D3D1x HAL
 D3D11 D3D10.1
FEATURE_LEVEL_10_0
FEATURE_LEVEL_9_3
FEATURE_LEVEL_9_1

HAL

GfxShaderMaker '-

featurelevel'

GfxShaderMaker.exe -platform D3D1x -featurelevel
FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1

FEATURE_LEVEL9_3

InitHAL

FEATURE_LEVEL_9_3

FEATURE_LEVEL_9_1